All Student Hours will be at MLC from now on. No zoom student hours.

Please let me know if this is a problem.

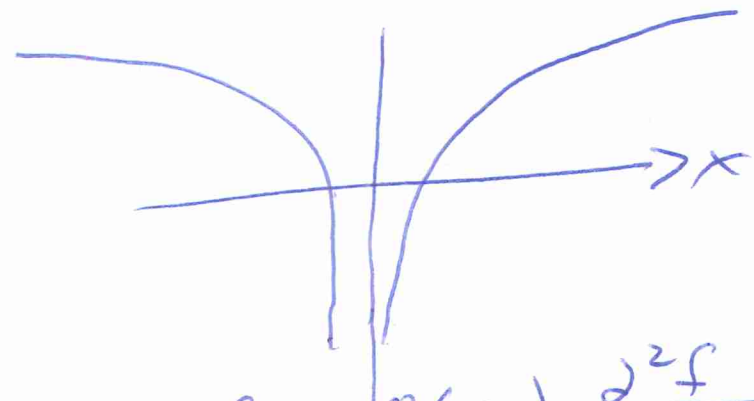
domain = open + continuous

<u>Def'n</u>

$C^0(\Omega)$ = set of all functions that are continuous

$= \{ f(x) : f(x) \text{ is cont. over domain } \Omega \}$

$C^1(\Omega) = \{ f(x) : f \in C^0(\Omega) \text{ and } \frac{\partial f}{\partial x_i} \in C^0(\Omega) \}$



<u>ex)</u> $y = \ln|x|$

$y \in C^0(-\infty, \infty)$? No

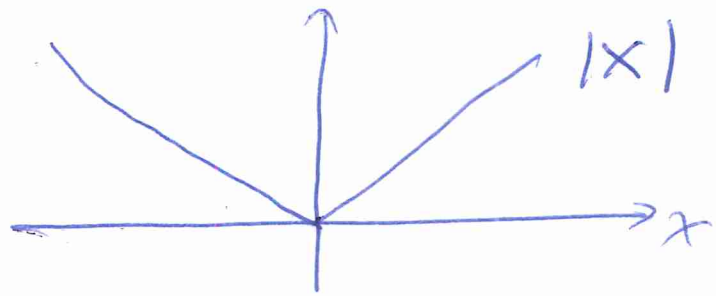
$y \in C^0(0, \infty)$? Yes

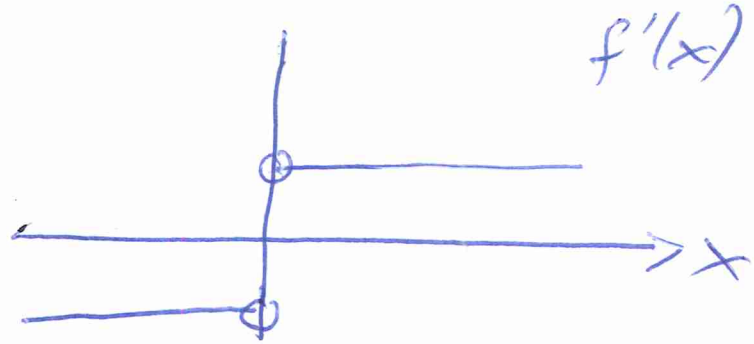
$C^2(\Omega) = \{ f(x) : f \in C^0(\Omega), \frac{\partial f}{\partial x_i} \in C^0(\Omega), \frac{\partial^2 f}{\partial x_i \partial x_j} \in C^0(\Omega) \}$

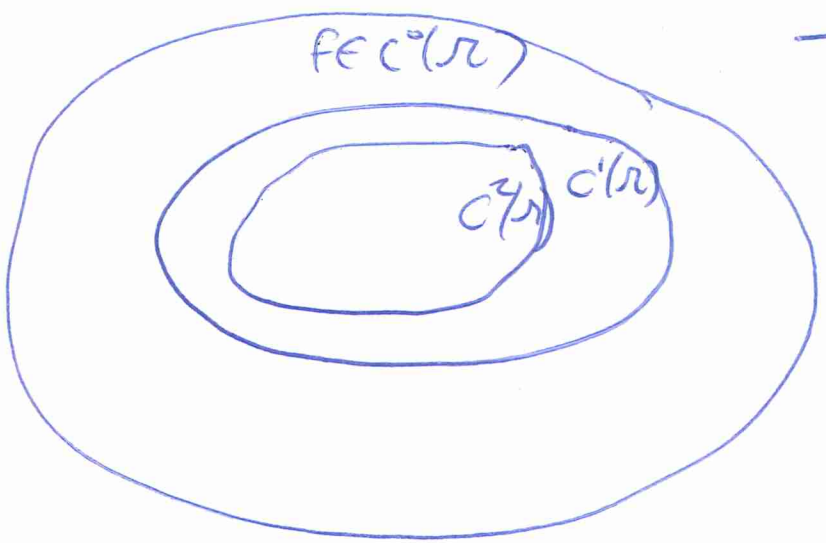
$C^\infty(\Omega) = \{ f(x) \in C^0(\Omega)$ and all partial deriv are $C^0(\Omega) \}$

ex) $f(x) \in C^0(\mathbb{R})$

$f'(x) \notin C^1(\mathbb{R})$







ex) $f(x) \in C^\infty(\mathbb{R})$

e^x, $\sin(x)$, $a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$

$f(x,y) = x^{4/3} y \qquad \in C^0(\mathbb{R}^2), \ \in C^1(\mathbb{R}^2)?$

$\in C^2(\mathbb{R}^2)? \quad \in C^\infty(\mathbb{R}^2)?$

$f \in C^0(\mathbb{R}^2)$

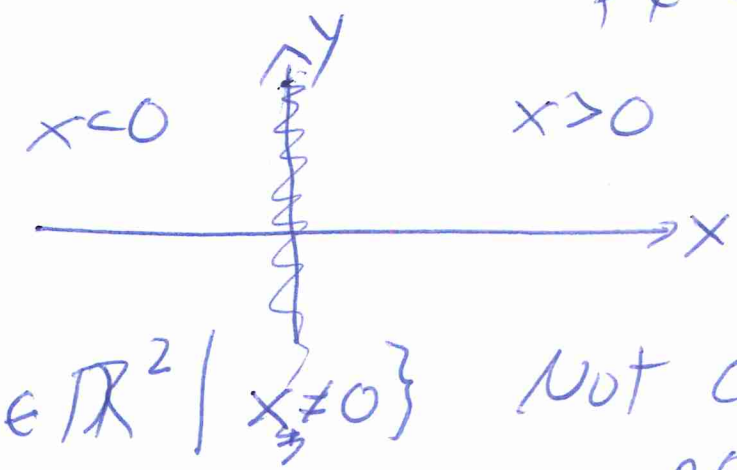
$\frac{\partial f}{\partial x} = \frac{4}{3} x^{1/3} y \qquad \frac{\partial f}{\partial y} = x^{4/3} \qquad f \in C^1(\mathbb{R}^2)$

$\frac{\partial^2 f}{\partial x^2} = \frac{4}{9} x^{-2/3} y \qquad \frac{\partial^2 f}{\partial x \partial y} = \frac{4}{3} x^{1/3} \qquad \frac{\partial^2 f}{\partial y^2} = 0$

$f \notin C^2(\mathbb{R}^2)$

$= \frac{4}{9} \frac{y}{x^{2/3}}$

$x \neq 0$



$\Omega = \{ \underline{x} \in \mathbb{R}^2 \mid x \neq 0 \}$ Not connected so not a domain.

$\Omega = \{ (x,y) \in \mathbb{R}^2 \mid x > 0 \}$

or $\Omega = \{ (x,y) \in \mathbb{R}^2 \mid x < 0 \}$ $\quad f(\Omega) \in C^\infty(\Omega)$

51.2

Surfaces and implicit fn Thm

Suppose $F(x,y,z) \in C^1(\Omega)$ $\Omega \subset \mathbb{R}^3$

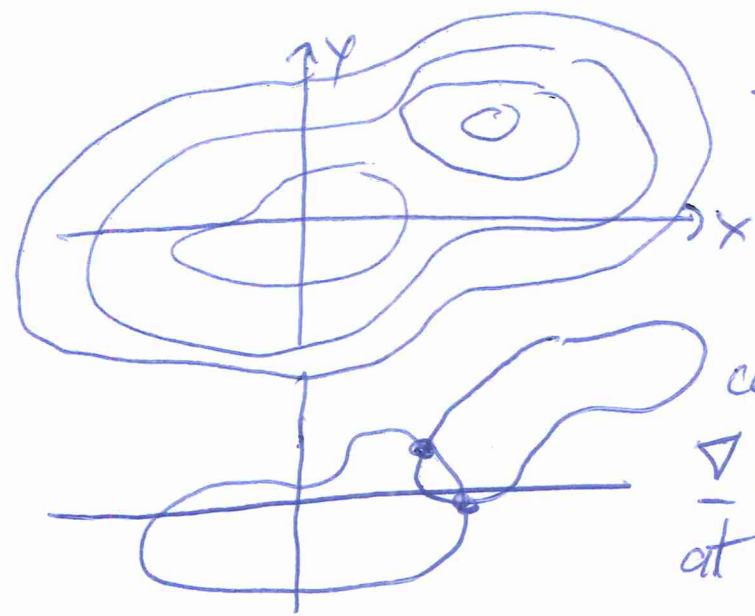
$$\underline{\nabla} F = \left\langle \frac{\partial F}{\partial x}, \frac{\partial F}{\partial y}, \frac{\partial F}{\partial z} \right\rangle$$

<u>If</u> $\underline{\nabla} F \neq (0,0,0)$ for all $(x,y,z) \in \Omega$

<u>Then</u> $F(x,y,z) = C$ is a level surface and each $(x_0, y_0, z_0) \in \Omega$, belongs to only one level surface.

In $\Omega \subset \mathbb{R}^2$,

Then level surfaces $\Rightarrow$ level curves



$\underline{\nabla} F \neq \underline{0}$ in Ω

could happen $\underline{\nabla} F = \underline{0}$ at a pt

F is not a fn. $\Leftarrow$

ex) $x^2+y^2=F(x,y)$

$\underline{\nabla} F = \langle 2x, 2y \rangle \neq \langle 0,0 \rangle$ except $(0,0)$

Level surfaces $x^2+y^2=C$

$C = 1, 2, 3, 4$

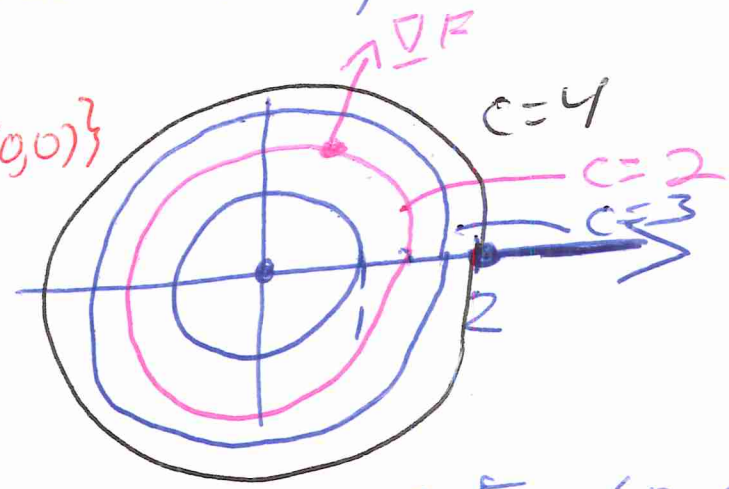
Change domain:

$\Omega = \{(x,y) \in \mathbb{R}^2 : (x,y) \neq (0,0)\}$

This is open & connected.

$\underline{\nabla} F \neq \langle 0,0 \rangle$ in this domain.

Level curves do not intersect in this domain. ☺



$(2,0)$

$\underline{\nabla} F\big|_{(2,0)} = \langle 4, 0 \rangle$

What happens when $\underline{\nabla} F = (0,0)$?

not a level "curve" (point)

Property of $\underline{\nabla} F$?

① $\underline{\nabla} F$ is a vector $\perp$ to $F(x,y,z)=C$

$y = f(x)$ $\quad$ $f'(x)$ is the slope of the tangent line

$y = f(x)$ is the level curve of $F(x,y)$

$F(x,y) = y - f(x) = 0$

$\underline{\nabla} F = \langle -f'(x), 1 \rangle$

$m = \frac{f'(x)}{1} = \frac{\text{rise}}{\text{run}}$

$\langle 1, f'(x) \rangle$

<u>Q</u> when can $F(x,y,z) = 0$ represent a <u>function</u> $z = f(x,y)$

ex) $z\cos(x+z) - y = \ln|z|$ cannot solve for z.

But is $z = f(x,y)$?

Implicit Function Thm

Let $F \in C^1(\mathbb{R}^3)$

If $\frac{\partial F}{\partial z}(x_0, y_0, z_0) \neq 0$ Then

$F(x,y,z) = C$ "can be solved for z" in terms of x and y in a neighborhood of (x_0, y_0, z_0), i.e. $z = f(x,y)$

open set that contains (x_0, y_0, z_0)

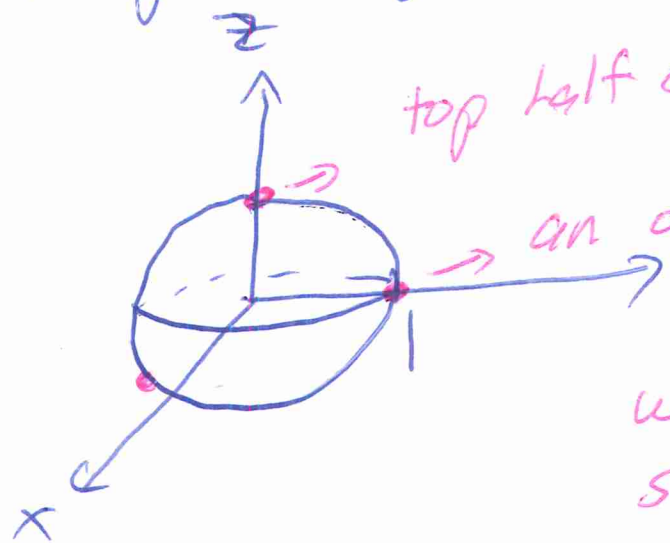
$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z} \qquad \frac{\partial z}{\partial y} = -\frac{F_y}{F_z}$$

$$F_x = \frac{\partial F}{\partial x}$$

ex) $x^2+y^2+z^2=1$ sphere

$F(x,y,z)=x^2+y^2+z^2 \qquad c=1$

$\frac{\partial F}{\partial z}=2z \qquad z=0$ is probably a problem



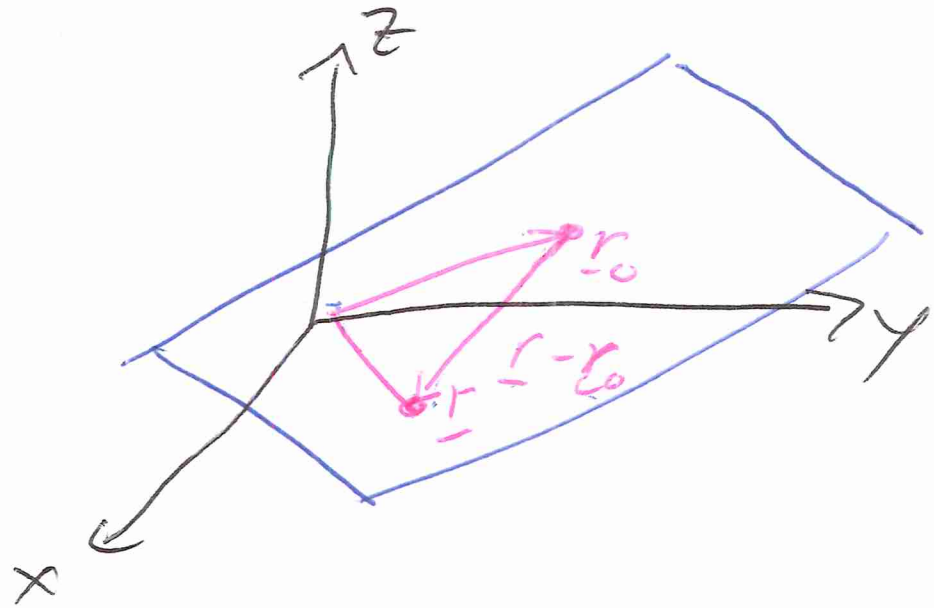
top half of sphere

an open set about $(0,1,0)$ where the sphere is a fn?

eq'n <u>eq'n of plane</u>

$\underline{r}_0$ = fixed pt on the ~~line~~ plane $\qquad \underline{n}$ = normal to plane

$\underline{r} = (x, y, z)$ be any other pt on plane

$\underline{r} - \underline{r}_0 =$ vector in plane

$$(\underline{r} - \underline{r}_0) \cdot \underline{n} = \underline{0}$$